THE PROJECTION FORMULA FOR A PROPER MORPHISM BETWEEN QUASI-COMPACT REGULAR SCHEMES WITH AFFINE DIAGONAL

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FFA (sources/stacks-chow/pinned-original-source.tex line 13913).

Allowed standard prerequisites (editorial).

- A morphism between quasi-compact regular schemes with affine diagonal is local complete intersection. More on Morphisms, Lemma labelled more-morphisms-lemma-morphism-regular-schemes-is-lci, cited at proof lines 13928-13930. Gives the first assertion and makes f^! available.
- The value of the Gysin map on the fundamental class. Same chapter, Lemma labelled lemma-lci-gysin-easy, cited at proof lines 13932-13934. Supplies f^![Y] = [X].
- The intersection product on regular schemes via the Chern character. Same chapter, Section labelled section-intersection-regular, invoked at proof lines 13934-13940 and 13958-13960. Writes the Chow classes as Chern characters capped against fundamental classes.
- Commutation of the Gysin map and centrality of Chern classes. Same chapter, Lemma labelled lemma-lci-gysin-commutes, cited at proof lines 13950-13954, and the centrality of Chern classes invoked at proof lines 13954-13958. Supplies the two middle equalities of the chain.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Same-chapter pointer into the author environment situation at source line 1372; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original section 0FEX: sources/stacks-chow/pinned-original-source.tex, source lines 13290–13290. Same-chapter pointer into the author environment lemma

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at source line 13290; invoked at proof line 13936 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0E9K: sources/stacks-chow/tags.txt, source lines 17855–17855. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 13931 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FF7: sources/stacks-chow/pinned-original-source.tex, source lines 13643–13643. Same-chapter pointer into the author environment lemma at source line 13643; invoked at proof line 13932 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FF6: sources/stacks-chow/pinned-original-source.tex, source lines 13613–13613. Same-chapter pointer into the author environment lemma at source line 13613; invoked at proof line 13954 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let $f : X \to Y$ be a morphism of schemes locally of finite type over S such that both X and Y are quasi-compact, regular, have affine diagonal, and finite dimension. Then f is a local complete intersection morphism. Assume moreover the Gysin map exists for f and that f is proper. Then*

$$f_*(\alpha \cdot f^!\beta) = f_*\alpha \cdot \beta$$

in $\mathrm{CH}^*(Y) \otimes \mathbf{Q}$ *where the intersection product is as in Section 0FEX.*

Proof. The first statement follows from More on Morphisms, Lemma 0E9K. Observe that $f^![Y] = [X]$, see Lemma 0FF7. Write $\alpha = ch(\alpha') \cap [X]$ and $\beta = ch(\beta') \cap [Y]$ $\alpha' \in K_0(\mathit{Vect}(X)) \otimes \mathbf{Q}$ and $\beta' \in K_0(\mathit{Vect}(Y)) \otimes \mathbf{Q}$ as in Section 0FEX. Set $c = ch(\alpha')$ and $c' = ch(\beta')$. We have

$$\begin{aligned} f_*(\alpha \cdot f^!\beta) &= f_*(c \cap f^!(c' \cap [Y]_e)) \\ &= f_*(c \cap c' \cap f^![Y]_e) \\ &= f_*(c \cap c' \cap [X]_d) \\ &= f_*(c' \cap c \cap [X]_d) \\ &= c' \cap f_*(c \cap [X]_d) \\ &= \beta \cdot f_*(\alpha) \end{aligned}$$

The first equality by the construction of the intersection product. By Lemma 0FF6 we know that $f^!$ commutes with c'. The fact that Chern classes are in the center of the bivariant ring justifies switching the order of capping $[X]$ with c and c'. Commuting c' with f_* is allowed as c' is a bivariant class. The final equality is again the construction of the intersection product. □